

Professional way to prepare programming contest problems

It seems you have exceeded 50 percents of CPU quota of invokers. Probably, checker/validator/generator/interactor or some solution is too slow or even hangs. If you exceed 100 percents you will not be able to run invocations or load invokers in any other way in about 5 minutes.

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Guess the Square

time limit per test: 1 second

memory limit per test: 256 megabytes

Keep up the Enthusiasm

— [twilighTTanvir](#)

This is an interactive problem.

In this problem, you have a very easy task. There is a square having a side length of a units ($1 \leq a \leq 10^5$). You have to determine its area, which is calculated as a^2 .

However, you are not given the value of a . To find it, you can ask at most 50 queries. In each query, you provide an integer x ($1 \leq x \leq 10^5$). The interactor will return true if $(a \oplus x) \geq a$ and false otherwise, where $\oplus$ denotes the bitwise XOR operation.

Find the area of the square.

Input

The first line of the input contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

Problem: **square-interactive**, id=525120 
 AI tips: Enabled (done)
 Owner: Tanvir Ahmed (tanvir_005)
 Note:  [click here to add note](#)
 Statements: [english](#)
 Checker: [checker.cpp](#) (1)
 Interactor: [interactor.cpp](#)
 Validator: [val.cpp](#) (1)
 Tests: [tests](#) (20)
 Solutions: [tanvir.cpp](#) (3/3)
 Package: for revision 9
 Verification: ([start](#))

[Review problem](#)

Access: OWNER, Write:3, Read:1
 Revision: 9 / 9
 Invokers: 48 / 60

[View changes](#)

https://polygon.codeforce..._005/square-interactive 

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For each test case, there is no initial input provided. You will receive the strings true, false, or -1 from the interactor as a response to your queries.

Output

For each test case, output your queries in the format $? x$ ($1 \leq x \leq 10^5$) and the final answer in the format $! \text{ area}$ on separate lines.

Do not forget to flush the output after each line.

Interaction

The first line of input contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

For each test case, the interactor will not provide any initial input. You must begin the interaction by sending your first query. To ask a query, output a single line in the format:

$? x$ ($1 \leq x \leq 10^5$)

In response to each valid query, you will receive a string from the standard input:

- true — if the condition $(a \oplus x) \geq a$ holds.
- false — otherwise.

If you make an invalid query or exceed the limit of 50 queries, you will receive -1 as a response. If you receive -1, your program must terminate immediately to receive a Wrong Answer verdict.

When you are ready to provide the final answer, output a single line in the format:

$! \text{ area}$

where area is the value a^2 . Providing the answer does not count as a query. After printing the answer, proceed to the next test case or terminate if it was the last one.

After printing any query or the answer, do not forget to output a newline and **flush the output buffer**. Otherwise, you will receive an Idleness Limit Exceeded verdict. To flush, use:



- `fflush(stdout)` or `cout.flush()` in C++;
- `System.out.flush()` in Java;
- `stdout.flush()` in Python;
- `flush(output)` in Pascal.

The interactor is not adaptive; the value of a is fixed for each test case before the interaction begins.

Example

input	
1	
false	
true	
true	
false	
false	
output	
? 1	
? 2	
? 3	
? 4	
? 5	
! 25	

Note

In the example, the hidden side length is $a = 5$.

- The first query is: ? 1 The result of $5 \oplus 1$ is 4. Since $4 < 5$, the interactor returns false.
- The second query is: ? 2 The result of $5 \oplus 2$ is 7. Since $7 \geq 5$, the interactor returns true.
- The third query is: ? 3 The result of $5 \oplus 3$ is 6. Since $6 \geq 5$, the interactor returns true.
- The fourth query is: ? 4 The result of $5 \oplus 4$ is 1. Since $1 < 5$, the interactor returns false.
- The fifth query is: ? 5 The result of $5 \oplus 5$ is 0. Since $0 < 5$, the interactor returns false.
- Finally, the area $5^2 = 25$ is reported.

Remember, you can ask different queries to determine the final answer.